

Project 1: Backgammon (board game)

Graphics, Simulation, Logic

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Programming Basics 2023/2024

do you have any questions or comments?



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Why?

To acquire the following knowledge or skills:

- basics of programming in C/C++
- basics of text graphics
- file operations
- performing a simple computer simulation
- implementation of player logic
- writing down the results of experiments in the form of a report¹

¹Extra points for completing an additional task

How? (technologies)

- You can use any ANSI/ISO compliant C/C++ compiler (Linux, MacOS, Windows).
- You can use one of two text graphics libraries: `conio.h` or `ncurses.h`
- You can use LaTeX or another text editor to write the report. Follow the pattern (PDF).²
- Tests can be performed on your private machine or on any other machine. It is important to describe its parameters precisely.²

²Applies to additional task

How? (C/C++)

- procedural programming using functions and structures
- use standard C or C++ libraries
- using data structures (array, stack, queue)
- generating random numbers (`srand()`, `rand()`)
- file manipulation using the `FILE` structure and C functions that operate on `FILE`, e.g., `fopen`, `fread`, `fwrite`, `ferr`, `fclose`

How? (limitations)

- It is prohibited to use object-oriented programming techniques (C++), in particular classes, inheritance mechanisms, polymorphism, overloading of functions and operators, etc. It is allowed to use data structures without methods (functions) inside the structures (C).
- It is forbidden to use global variables, except for variables of simple types (e.g. int, float, etc., without pointers) preceded by the `const` specifier.
- The use of the C++ template library is prohibited.
- The length limit (number of characters) of any function used (including `main()`) is 555 bytes.

Note: Failure to comply with the above restrictions will result in disqualification of the project and obtaining 0 points.

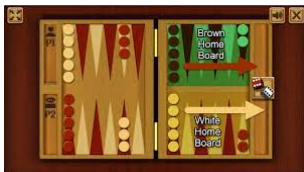
What?

- Obligatory (graphics and simple logic, max. 6 pts)
 - 1 Basic user interface (1 point)
 - 2 Visualization of position in the game *backgammon* (1 pt)
 - 3 Saving the game state to a file and loading the game from the file (2 pts)
 - 4 Playing the next game move from the loaded game state (2 pts)
- Optional (simulation and gameplay logic, max. 9 pts)³
 - 1 Implementation of the rule of capturing the opponent's pawn (1 pt)
 - 2 Implementation of the rule of removing pawns (1 pt)
 - 3 Game simulation from any position to the end of the game (2 pts)
 - 4 Save the game in the form of subsequent moves from the beginning to its end (1 pt)
 - 5 Visualization of the gameplay based on the loaded save (2 pts)
 - 6 Complete UI (1 pt)
 - 7 Ranking of players identified by name (*Hall of Fame*) (1 pt)
- Additional (computer player logic, max. 3 points)
 - Own proposal for computer player logic
 - Comparison of different concepts of computer gamer logic
 - Results report

³ Scored only any *prefix* from all task list numbers, i.e. a failure to complete task number i results in not scoring subsequent tasks $(i + 1, \dots)$

Introduction

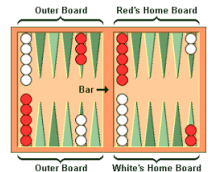
- Game name: *backgammon*, *nardy*
- A board game for two players.
- It involves moving pawns according to the number of points rolled on a board consisting of 24 fields.
- The winner is the person who first removes all his pawns from the board.
- This is a logical (strategic) game.



<https://en.wikipedia.org/wiki/Backgammon>

Game Hardware

- BOARD: it consists of four parts, each of which has 6 triangle-shaped fields drawn.
- PAWNS: the game requires 15 white pawns and 15 red pawns (black, brown).
- DICE: 2 (classic) dice



Basic rules

- The board is divided into several parts: red *base* (*home board*) (fields 1-6), outer board fields (7-18) *outer board*), base (home) of white player (squares 19-24).
- The vertical strip in the middle is called *bar*.
- The vertical strip on the right side of the board is called the *court*.
- The aim of the game is to place all your pawns in your base and then remove them from the board (outside) to the court.
- The winner is the player who first gets rid of all his pawns from the board.

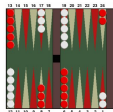


Figure: Starting arrangement of pawns.

Starting the game

- The game begins with the appropriate arrangement of pawns on the board.
- Red player moves pawns clockwise, white player moves pawns counterclockwise.
- At the beginning, players roll one dice and the one who rolls the higher number starts the game.
- Then he rolls two dice and moves his pawns the appropriate number of spaces.

Moving pawns

- Each of the two dice allows one of the pawns to move (we move the pawn by a number of spaces equal to the number of spots on the dice).
- You can move one pawn by the sum of the two dice, as long as at least one of the intermediate spaces (from one dice) is free.
- A player can only move a pawn to a square that is empty, has only his own pawns on it, or only one opponent's pawn on it.
- You cannot place a pawn on a square that contains more than one opponent's pawn.
- If a player cannot make a move corresponding to one of the dice, he does not make it.
- If a player cannot make any move, he loses his turn.

Double

- Rolling the same value on both dice is called *double*.
- When both dice roll the same number (marked a), the player can make four moves worth a .
- For example: if we roll two points on each dice, we can: move one pawn by eight squares, or two pawns by four squares, or two pawns by two squares and one by four squares, or four pawns, each by two squares .

Capturing pawns

- Capturing pawns occurs when a player's pawn is moved to a field containing one of the opponent's pawns.
- The opponent's captured pawn is moved to the bar.
- The player whose pawn is on the bar cannot move with the pawns on the board (fields 1-24) until he removes all the pawns from the bar.

Removing pawns from the bar

- The pawns from the bar are moved to the opponent's base.
- They are placed on an empty square or on a square containing one opponent's pawn.
- After rolling two dice, one of the dice is selected, with the number of spots equal to the distance of the field from the manor.
- If the opponent has at least two pawns on each field in his base, he is not able to introduce pawns from the bar, so the player loses his turn.
- Similarly, if both squares that indicate the dice are owned by the opponent (i.e. he has at least two pawns on each of these squares).

Removing pawns from the board

- Removing pawns from the board is possible when you have all unremoved pawns in your base.
- If a player has one pawn out of the base and the rest in the base, then if moving one die is enough to get that pawn into the base, then the other die can move one of the pawns out of the base and into the court.
- Pawns can be moved out of the base if they are on squares that are within the number of points rolled from the manor.
- If the number of points on the dice is greater than the distance of the furthest pawn from the base to the court, the player removes his furthest pawn.
- Once a pawn is removed from the board, it does not return to the board.

Scoring rules

- Players can play not only one duel, but entire matches (series of games).
- One point is scored for each victory.
- If a player brings all of his pieces into the court before his opponent manages to get even one of his pieces into the court, then the player gets an additional point.
- However, if the opponent has at least one pawn on the board, the winner receives three points for winning.
- In the basic variant (not considered within the project), players play using the so-called *double dice*. Using such a dice means multiplying the number of points you get for a victory by the value indicated on the dice (2, 4, ... 64).

Comment

- Basic user interface
 - Information about the author (name, surname, index number)
 - Possibility to select command in *menu*
 - Applying text graphics
- Visualization of position in the game (example)



- Playing the next game move from the loaded game state
 - Human movement selection.
 - It is necessary to simulate dice rolling and indicate possible movements for a human or check their correctness after making a move.
 - The implementation of capturing a pawn, removing a pawn from the house to the court or bringing a pawn from the board to the board is not necessary.

Comment

- Implementation of the rule of capturing the opponent's pawn
 - The player is forced to play this way and the system should control it.
 - The captured pawn should be the opponent's pawn closest to the court.
- Implementation of the rule of removing pawns
 - The player is forced to remove maximum possible number of pawns.
- Game simulation from any position to the end of the game
 - Implementation of possible moves and control of all game rules.
 - In the simulation, both players are controlled by a human, but *may* be limited by previously indicated rules or other rules.
- Save subsequent moves for the whole game
 - Save the state to a text file. Binary files are not allowed.
- Visualization of the gameplay based on the saved file
 - Forward and backward functionality (by one move)
 - Go to the beginning or to the end of the game

Comment

- Own proposal for a computer player logic
 - Use a simple rules system for a computer player with elements of randomness.
 - Consider the system for evaluating positions and movements.
 - Only limited depth search due to the time restrictions.
- Comparison of computer player logic
 - Implementation of several variants of the strategy (parameterized) for playing the game by a computer player.
 - Conducting matches (series of games) in a round-robin system (multiple times).
 - Demonstration of test results.
- Results report
 - A detailed template with the required report details will be provided in a separate file.